

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12402230
Kind Code	B1
Date of Patent	August 26, 2025
Inventor(s)	Law; Calvin

Apparatus and method of vehicle headlamp control device

Abstract

A pattern-light lamp comprises a lamp unit, a control box connected to the lamp unit via a wired or wireless method, a mobile user interface connected wirelessly to the control box, and controlling the control box, which in turn controls the lamp unit. The lamp unit comprises a modifiable light rate component, a light pattern component, and a light color component. The control box comprises a modifiable light rate control unit, a light pattern control unit, and a light color control unit. The mobile user interface comprises a modifiable light rate module, a light pattern module, and a light color module.

Inventors:	Law; Calvin (Chino, CA)
Applicant:	Law; Calvin (Chino, CA)
Family ID:	1000007524772
Appl. No.:	18/518504
Filed:	November 23, 2023

Publication Classification

Int. Cl.: H05B47/19 (20200101)

U.S. Cl.:

CPC H05B47/19 (20200101);

Field of Classification Search

CPC: H05B (47/19)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9374874	12/2015	Ewing	N/A	G08C 17/02
10605427	12/2019	Chen	N/A	F21S 41/43
11678296	12/2022	Luterman	370/329	H04W 76/40
12160943	12/2023	Law	N/A	B60Q 1/04
D1065628	12/2024	Sperling	D26/28	N/A
2019/0118699	12/2018	Wu	N/A	B60Q 1/076
2020/0331552	12/2019	Horn	N/A	B62J 6/024
2023/0272895	12/2022	Yonemaru	348/148	F21S 41/13

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3043822	12/2018	CA	H05B 47/105
2024121133	12/2023	JP	N/A
2024130011	12/2023	JP	N/A
2025019877	12/2024	JP	N/A
WO-2021042002	12/2020	WO	B60H 1/00785

Primary Examiner: Houston; Adam D

Background/Summary

FIELD OF INVENTION

(1) The present invention is a smart headlamp lighting control system that allows the user to modify the color, flash rate, and lighting pattern of the various lights within aftermarket headlamps. The headlamp uses a control box (or circuit board) and includes a unique user interface. The user interface comes in the form of a mobile application that allows the user to choose the color as well as the light-up or flashing pattern (e.g., sequential light up), a user-devised schedule for color, rate, and pattern for each of the various lights within the headlamp, which can be based upon music. It is the first of its kind to provide a solution both for the control box and user interface.

BACKGROUND

(2) Aftermarket equipment, or the secondary market of the automotive industry, is concerned with the manufacturing, remanufacturing, distribution, retailing, and installation of all vehicle parts, chemicals, equipment, and accessories, after the sale of the automobile by the original equipment manufacturer (OEM) to the consumer. The purchase of such aftermarket parts is a popular way for automobile enthusiasts to personalize their vehicles.

(3) One such popular piece of equipment is the headlamp, which is a lamp attached to the front of a vehicle to illuminate the road ahead. While the color of the lights within the lamp, from the running and primary lights (white or selective yellow) and blinkers (amber), are subject to regulation, there is still room for some adaption in order to permit vehicle personalization. However, it has heretofore been a complicated process for the user to modify and manage such personalization.

(4) Thus, it is desirable for a solution both for the control box appended to the headlamp and a mobile user interface to customize the headlamp.

OBJECTIVE OF THE INVENTION

(5) It is an object of this invention to create a solution for the control box appended to the headlamp in order to regulate color, rate, and pattern of each light therein.

(6) It is an additional object of this invention to provide a user-friendly mobile interface for the user

to easily modify and manage the color, rate, and pattern of each light.

(7) It is a further object of this invention to prevent the user from violating any vehicle lighting laws while in motion.

(8) It is a further object of this invention to provide music data, which commands the color, rate, and pattern of the headlamp according to a pre-determined algorithm.

Description

SUMMARY OF THE INVENTION

Brief Description

(1) The present invention will be understood and appreciated more fully from the following detailed description taken in conjunction with the drawings in which:

(2) FIG. 1 illustrates a block diagram of the three components of the invention: headlamp, control box, and mobile user interface.

(3) FIG. 2 illustrates a block diagram of the modifiable portions of each of the three components of the invention.

(4) FIG. 3 illustrates a block diagram of the mobile user interface coupled with the software download module.

(5) FIG. 4 illustrates a block diagram showing the GPS module introduced to the mobile user interface.

(6) FIG. 5 illustrates a block diagram showing an embodiment of the invention using an on-board diagnostics connector.

(7) FIG. 6 illustrates the headlamp component of the invention and the various types of lights therein for which the color may be modified separately.

(8) FIG. 7 illustrates the headlamp component of the invention and the various types of lights therein for which the light up pattern may be modified separately.

(9) FIG. 8 illustrates the headlamp component of the invention and the various types of lights therein for which the light rate may be modified separately.

(10) FIG. 9 illustrates a sequence program module in the mobile user interface.

(11) FIG. 10 illustrates a music synchronizing program connected to the mobile user interface.

DETAILED DESCRIPTION

(12) The invention will be described in the context of a preferred embodiment.

(13) FIG. 1 illustrates a block diagram showing how the three components of the invention connect and interact. The headlamp unit **101** connects to the control box **102**, which connection can be wired or wireless (in this embodiment it is wired). The control box **102** then connects wirelessly through Bluetooth or Wi-Fi technology to the mobile user interface **103**, which in the preferred embodiment exists in the form of an application downloaded onto the user's mobile phone **104**. The control box **102** gives instruction from the mobile **104** user interface **103** to the headlamp **101**.

(14) FIG. 2 illustrates a block diagram showing the modifiable portions of each of the three components of the invention. The headlamp **204** contains modifiable light rate **201** (how it flashes, e.g. 1× per second, 10× per second), light pattern **202** (e.g., flash or sequential light-up), and light color **203**, the instructions for which come from the control box **205**. The control box **205** contains modifiable light rate **206**, light pattern **207**, and light color **208**, the instructions for which come from the mobile user interface **209**. The mobile user interface **209** contains modifiable light rate **210**, light pattern **211**, and light color **212**.

(15) FIG. 3 illustrates a block diagram showing the mobile user interface **310** coupled with a software **314** download module **315**, which permits the user to acquire new colors **313**, light rates **311**, and light patterns **312**. The control box **305** also contains a new software module **309** for additional rate **306**, pattern **307**, and color **308** options as they come available. The headlamp unit

301 is again shown with modifiable light rate **302**, light pattern **303**, and light color **304**.

(16) FIG. **4** illustrates a block diagram which introduces a global positioning software (GPS) module **413** into the mobile user interface **409**. The GPS **413** prevents the user from violating any vehicle lighting laws by prohibiting any potentially illegal combinations of light rate **410**, color **411**, and pattern **412** while the vehicle is moving. The mobile user interface is shown connecting to the control box **405**, which itself contains light rate **406**, color **407**, and pattern **408**. The headlamp unit **301** is again shown with modifiable light rate **402**, light color **403**, and light pattern **404**.

(17) FIG. **5** illustrates a block diagram of a second embodiment of the invention, for use when a user does not wish to use their mobile phone, hence an alternate to the GPS on the mobile unit is needed. The headlamp **501** is shown with its components of modifiable light rate **502**, pattern **503**, and color **504**. The control box **505** is shown connecting to the headlamp **501**, again with its components of modifiable light rate **506**, pattern **507**, and color **508**. An on-board diagnostics module (OBD) **513**, which plugs into the vehicle under the dashboard, with an OBD reader **514** that connects to the control box **505**, again shown with light rate **506**, pattern **507**, and color **508**. The control box **505** can have a wired connection to the OBD **513**, which is an alternative way to prevent the user from violating any vehicle lighting laws by prohibiting any potentially illegal combinations of light rate **506**, color **508**, and pattern **507** while the vehicle is moving, turning, inter alia. In yet another embodiment, the mobile user interface **509** is shown connected to the OBD reader **514** as well as the control box **505**, again containing modifiable light rate **510**, color **511**, and pattern **512**. In one embodiment, the apparatus can have the GPS included within the control box **505**.

(18) FIG. **6** illustrates the headlamp component **607** comprising an upper daylight running light **601**, lower daylight running light **602**, upper primary lights **603**, lower primary lights **604**, and turn signal lights **605**. The color **606** of each of these lights may be modified by the user.

(19) FIG. **7** illustrates the headlamp component () comprising an upper daylight running light **701**, lower daylight running light **702**, upper primary lights **703**, lower primary lights **704**, and turn signal lights **705**. The light up pattern **706** of each of these lights may be modified by the user.

(20) FIG. **8** illustrates the headlamp component **707** comprising an upper daylight running light **801**, lower daylight running light **802**, upper primary lights **803**, lower primary lights **804**, and turn signal lights **805**. The light rate **806** of each of these lights may be modified by the user.

(21) FIG. **9** illustrates a sequence program module **905** permitting the user to design a sequence of all three modifiers, color **901**, rate **902**, and pattern **903**, instead of changing each one at a time. The sequence program module **905** connect to the mobile user interface **904**.

(22) FIG. **10** illustrates a music synchronizing program **1013** downloaded to the mobile user interface **1009**, which itself has a microphone module **1015** and a music input module **1014**, which will use artificial intelligence to synchronize the light rate **1010**, color **1011**, and pattern with the beats of music coming through the music input module **1014**. The music input module **1014** is the form of data from the cloud or a memory card, etc. The microphone module **1015** records the analog music played in the environment, which it then converts to data. The mobile user interface then connects to the control box **1005**, which contains the rate **1006**, color **1007**, and pattern **1008** modules. The control box **1005** then controls the headlamp unit with rate **1002**, color **1003**, pattern **1004** modules.

Claims

1. A pattern-light lamp comprising: a lamp unit; a control box, wherein said lamp unit connects to said control box with at least one of wired or wireless method; a mobile user interface, wherein said mobile user interface connects wirelessly to said control box with at least one of Bluetooth or Wi-Fi technology; wherein said mobile user interface gives instructions to said control box, which in turn gives commands to said lamp unit; wherein said lamp unit comprises a modifiable light rate

component that controls how lights flash and flashing speed; wherein said lamp unit further comprises a light pattern component that controls light patterns; wherein said lamp unit further comprises a light color component that controls light colors; wherein said control box further comprises a modifiable light rate control unit that commands said light rate component; wherein said control box further comprises a light pattern control unit that commands said light pattern component; wherein said control box further comprises a light color control unit that commands said light color component; wherein said mobile user interface further comprises a modifiable light rate module that commands said modifiable light rate control unit; wherein said mobile user interface further comprises a light pattern module that commands said light pattern control unit; wherein said mobile user interface further comprises a light color module that commands said light color control unit.

2. The lamp of claim 1, wherein said mobile user interface further comprises a software download module that permits a user to acquire new light colors, new light color control unit, new light color module; new light rates, new light rate unit, new light rate module; and new light patterns, new light pattern unit, new light pattern module.

3. The lamp of claim 1, wherein said mobile user interface further comprises a global positioning software (GPS) module that gathers local traffic laws and builds dynamic filters that filter out illegal light patterns.

4. The lamp of claim 1, wherein said control box comprises a GPS component.

5. The lamp of claim 1 further comprises an on-board diagnostics module (OBD) that connects to a vehicle's internal system under a dashboard; wherein said OBD connects to said control box; wherein said OBD further comprises a reader module that provides data to said control box.

6. The lamp of claim 5, wherein said OBD connects to said mobile user interface.

7. The lamp of claim 1 further comprises upper and lower daylight running lights, upper and lower primary lights, and turn signal lights.

8. The lamp of claim 1, wherein said control box further comprises a sequence control unit that permits a user to design a sequence of all three control units: said light color control unit, light pattern control unit, and light rate control unit; wherein said sequence control unit connects to said mobile user interface.

9. The lamp of claim 1, wherein said mobile user interface further comprises a music synchronizing module that interfaces with said sequence control unit; wherein said music synchronizing module is downloadable; wherein said music synchronizing module in turn comprises a microphone module and a music input module that use artificial intelligence to synchronize a sequence of musical beats with a sequence of light rates, colors, and patterns.

10. The lamp of claim 9, wherein said music input module is a data stream from a digital storage, comprising at least one of a digital cloud, a memory card, or other digital storage device.

11. A method for a pattern-light lamp comprising: providing a lamp unit; providing a control box; connecting said lamp unit to said control box with at least one of wired or wireless method; providing a mobile user interface; connecting said mobile user interface wirelessly to said control box with at least one of Bluetooth or Wi-Fi technology; wherein said mobile user interface gives instructions to said control box, which in turn gives commands to said lamp unit; wherein said lamp unit comprises a modifiable light rate component that controls how lights flash and flashing speed; wherein said lamp unit further comprises a light pattern component that controls light patterns; wherein said lamp unit further comprises a light color component that controls light colors; wherein said control box further comprises a modifiable light rate control unit that commands said light rate component; wherein said control box further comprises a light pattern control unit that commands said light pattern component; wherein said control box further comprises a light color control unit that commands said light color component; wherein said mobile user interface further comprises a modifiable light rate module that commands said modifiable light rate control unit; wherein said mobile user interface further comprises a light

pattern module that commands said light pattern control unit; wherein said mobile user interface further comprises a light color module that commands said light color control unit.

12. The method of claim 11, wherein providing a software download module that permits a user to acquire new light colors, new light color control unit, new light color module; new light rates, new light rate unit, new light rate module; and new light patterns, new light pattern unit, new light pattern module.

13. The method of claim 11, wherein providing a global positioning software (GPS) module that gathers local traffic laws and builds dynamic filters that filter out illegal light patterns.

14. The method of claim 11, wherein said control box comprises a GPS component.

15. The method of claim 11, wherein providing an on-board diagnostics module (OBD) that connects to a vehicle's internal system under a dashboard; wherein said OBD connects to said control box; wherein said OBD further comprises a reader module that provides data to said control box.

16. The method of claim 11, wherein said OBD connects to said mobile user interface.

17. The method of claim 11, wherein said lamp unit further comprises upper and lower daylight running lights, upper and lower primary lights, and turn signal lights.

18. The method of claim 11, wherein providing a sequence control unit that permits a user to design a sequence of all three control units: said light color control unit, light pattern control unit, and light rate control unit; wherein said sequence control unit connects to said mobile user interface.

19. The method of claim 11, wherein providing a music synchronizing module that interfaces with said sequence control unit; wherein said music synchronizing module is downloadable; wherein said music synchronizing module in turn comprises a microphone module and a music input module that use artificial intelligence to synchronize a sequence of musical beats with a sequence of light rates, colors, and patterns.

20. The method of claim 19, wherein said music input module is a data stream from a digital storage, comprising at least one of a digital cloud, a memory card, or any digital storage device.
